

Jiaming Feng

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Research Interests

Medical image analysis; computer vision for image-guided intervention; preoperative-to-intraoperative registration; anatomical segmentation; augmented and mixed reality for surgery; surgical scene understanding

Education

University of Leeds

MSc Advanced Computer Science

– Awarded the International Excellence Scholarship.

Sep 2025 – Sep 2026 (expected)

Leeds, United Kingdom

Henan Agricultural University

B.Eng. in Computer Science and Technology

– Average grade: 81%; awarded Excellent Undergraduate Thesis.

Sep 2021 – Jun 2025

Henan, China

Research Experience

AI in Medicine and Surgery Group, University of Leeds

Research Collaboration under Dr Sharib Ali

Sep 2025 – Present

Leeds, United Kingdom

- Conduct research on computer vision for image-guided liver intervention, with a focus on preoperative-to-intraoperative fusion, 3D anatomical landmark segmentation, and geometry-aware registration for AR-guided surgery.
- Developed *GeoTranMesh*, a geometry-guided multi-branch mesh Transformer for 3D liver surface landmark segmentation, integrating hierarchical mesh encoding, directional feature fusion, and multi-task supervision; achieved 59.5% overall Dice and 4.4 mm Chamfer Distance on a liver mesh benchmark.
- Designed *Vis2Reg*, a visibility-aware, landmark-free geometric 3D–2D registration framework for laparoscopic liver surgery, using differentiable rasterization and mask-guided self-supervision for robust alignment under severe partial visibility.
- Currently developing robust video-based preoperative-to-intraoperative liver registration for dynamic surgical scenes, extending static geometric fusion towards temporally consistent intraoperative guidance.
- Collaborate with computer scientists and hepatobiliary surgeons on clinically grounded mixed-reality and image-guided intervention research.

Wheat Seedling Detection System for Precision Agricultural Monitoring

Henan Agricultural University, supervised by Prof. Xi Lei

Mar 2024 – Jul 2024

Henan, China

- Led the development of an end-to-end wheat seedling detection system for real-world field monitoring, supporting automated analysis of seedling distribution from drone-captured continuous video frames.
- Built a deployable computer vision system capable of identifying sparse and dense wheat seedling regions in farmland scenarios, contributing to practical wheat seedling inspection and crop growth monitoring.
- Deployed the system to a mobile platform and supported its real-world application in agricultural production, enabling on-site visual assessment and real-time interaction during field inspection workflows.
- Translated AI-based visual analysis into an operational agricultural application, demonstrating experience in developing usable systems beyond offline model experimentation and contributing to field-level agricultural decision support.

Publications

- **Feng, J.**, Zhang, X., Farid, S., & Ali, S. *GeoTranMesh: a geometry-guided multi-branch mesh transformer for 3D liver segmentation. International Journal of Computer Assisted Radiology and Surgery*, 2026. Accepted at *IPCAI 2026*. doi:10.1007/s11548-026-03600-8.
- **Feng, J.** *The application of emotion recognition from two types of physiological signals in human-machine interaction within autonomous driving. AIP Conference Proceedings*, 3194(1):030003, 2024. doi:10.1063/5.0223242.

Manuscript Under Review

- **Feng, J.**, Zhang, X., Farid, S., & Ali, S. *Vis2Reg: Visibility-Aware Landmark-Free Geometric 3D–2D Registration for Liver Laparoscopy*. Under review at *MICCAI 2026*.

Manuscript in Preparation

- **Feng, J.**, Ali, S., et al. *Preoperative-to-Intraoperative Robust Liver Registration in Dynamic Surgical Scenes*. Manuscript in preparation for submission to *IEEE Transactions on Medical Imaging*.

Industry Experience

RainbowWireless (Beijing) New Technology Co., Ltd.

Sep 2024 – Dec 2024

AI Algorithm Intern

Beijing, China

- Developed a retrieval-augmented diagnostic assistant for intelligent vehicle maintenance in collaboration with a major automotive enterprise in China, transforming large-scale maintenance manuals into a queryable knowledge system for fault diagnosis and repair support.
- Built pipelines for parsing, structuring, indexing, and retrieving knowledge from heterogeneous technical manuals containing tens of thousands of pages, improving document understanding and retrieval robustness in real-world industrial settings.
- Designed a symptom-driven diagnostic workflow that maps sparse user-reported fault symptoms to relevant fault localisation and repair procedures, supporting practical decision-making in complex maintenance scenarios.
- Gained experience in building AI systems that operate on incomplete evidence and large unstructured knowledge sources, which is conceptually relevant to data-driven support for high-stakes, real-world workflows.

Honours and Awards

- Second Prize, AI Innovation Competition, China Robotics and Artificial Intelligence Competition, National Finals, 2024.
- Third Prize, Asia and Pacific Mathematical Contest in Modeling, APMCM, 2024.
- Third Prize, ShuWei Cup International Mathematical Contest in Modeling, IMCM, 2024.
- Second Prize, CSEE Cup Electrical and Mathematical Modeling Competition, 2023.

Technical Skills

- **Research Skills:** medical image analysis, surgical computer vision, anatomical segmentation, 3D–2D registration, preoperative-to-intraoperative image fusion, self-supervised learning.
- **3D and Graphics Tools:** mesh and point-cloud processing, geometric deep learning, camera geometry, 3D model visualisation, Blender, Unity.
- **Frameworks and Tools:** PyTorch, Linux/Unix, Git, LaTeX, AI system deployment, mobile deployment.
- **Languages:** English (IELTS 6.5), Chinese (native).

References

- **Dr Sharib Ali**, Associate Professor, School of Computer Science, University of Leeds. Email: S.S.Ali@leeds.ac.uk.
- **Prof. Xi Lei**, Professor, Henan Agricultural University. Email: xil@henau.edu.cn.